

EXPERIMENT 9

Aim

To implement a **Recurrent Neural Network (LSTM)** to predict future values in a time series dataset.

Problem Statement

Time series forecasting is used to predict future values based on past data.

The problem is to predict the number of airline passengers for future time steps using historical monthly data.

Dataset Link

<https://www.kaggle.com/datasets/vedatgul/airline-passengers>

Methodology

Data Collection

The dataset was loaded and the relevant feature, passenger count, was selected for analysis.

Data Preprocessing

The data was normalized using MinMaxScaler to scale values between 0 and 1, which helps improve model performance and convergence.

Sequence Creation

The time series data was converted into sequences (time steps), where previous observations are used as input to predict future values. This is essential for LSTM models to learn temporal dependencies.

Data Splitting

The dataset was split into training and testing sets to evaluate model performance on unseen data.

LSTM Model Architecture

An LSTM model was built consisting of an LSTM layer to capture sequential patterns and a Dense output layer to generate predictions.

Model Compilation

The model was compiled using the Adam optimizer for efficient training and Mean Squared Error (MSE) as the loss function, suitable for regression tasks.

Model Training

The model was trained for multiple epochs, allowing it to learn patterns, trends, and seasonality in the data.

Prediction & Evaluation

The trained model was used to predict future passenger values, and the predictions were compared with actual values to evaluate performance and accuracy.

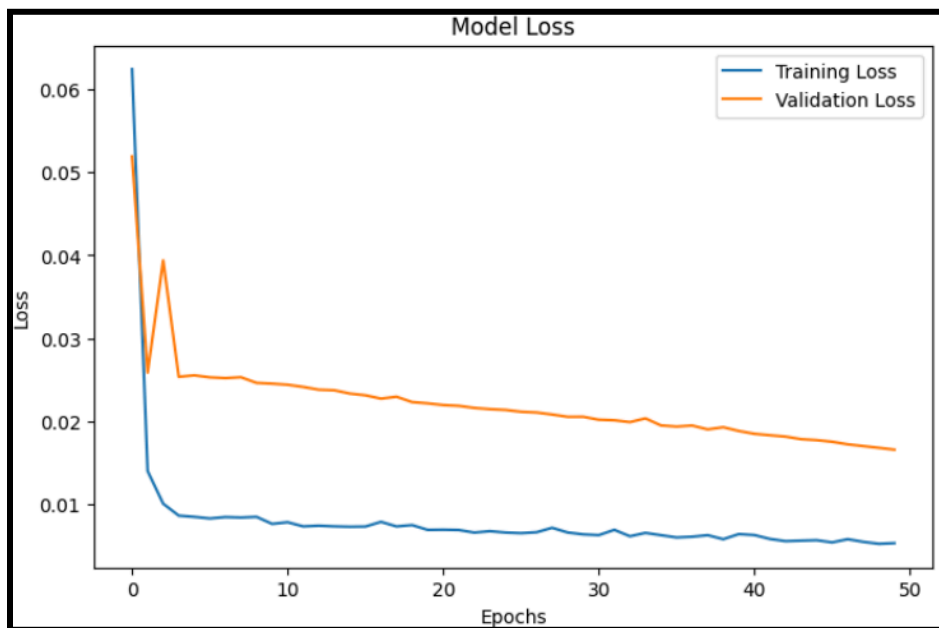
Result

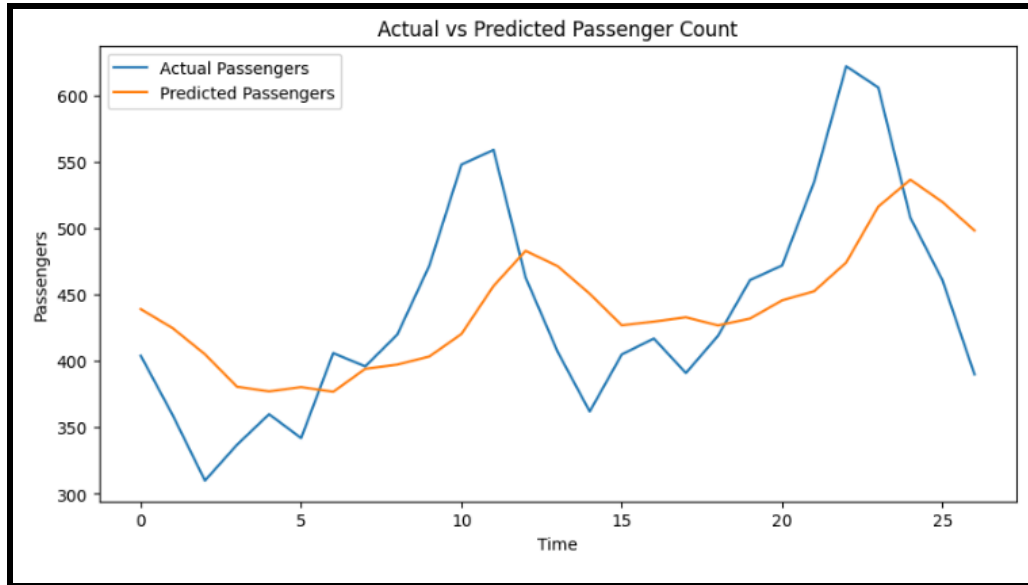
- **Loss (MSE):** Decreased over epochs
- Model successfully captured trend of passenger data

Performance Metrics:

- Mean Squared Error (MSE)

Model Graphs:





Conclusion

The LSTM model effectively captured time-dependent patterns in the dataset by utilizing its memory cells to retain past information over long sequences. This allowed the model to understand trends, seasonality, and dependencies in the data. As a result, it was able to predict future passenger trends with good accuracy.

Furthermore, the model adapted well to sequential data, making it highly suitable for time series forecasting tasks. Its ability to learn from historical patterns and make reliable future predictions makes LSTM a powerful technique for applications such as demand forecasting, stock price prediction, and traffic analysis.